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Project Title: Evaluating LLM Reasoning and Generation for Cancer Upstaging in Pathology Reports

Project Progress Report

Part 1a: Integration of Project Plan Feedback & Experimental Updates

The original project plan has been refined based on the feedback to better reflect the current state-of-the-art in natural language processing (NLP) and to accommodate strict hardware constraints (Google Colab T4 GPU). The following structural changes have been implemented:

- **Literature & Model Shift:** Feedback noted that standard Chain-of-Thought (CoT) prompting literature is evolving rapidly as modern models integrate reasoning natively. Consequently, the proposed use of Llama-3 was abandoned. The literature review and model selection have been updated to compare an instruction-tuned generalist model (Qwen-2.5-7B-Instruct) against a model specifically trained for complex logic and native reasoning (microsoft/phi-4).
- **Evaluation Metric Correction (The “Semantic Trap”):** Feedback highlighted the clinical limitations of BERTScore for evaluating medical rationales, as it may score logically flawed but semantically similar text highly. To address this, the experimental design now utilizes a dual-metric approach. A deterministic **Faithfulness Metric** (strict substring verification) has been engineered to flag exact-match clinical hallucinations. BERTScore has been retained strictly to serve as a comparative baseline, allowing the study to actively measure and expose instances where semantic similarity metrics falsely validate dangerous clinical errors that the Faithfulness Metric successfully catches.
- **Experimental Design Refinement:** To allow for rigorous evaluation without exceeding computational limits, the dataset scope was reduced. Rather than processing the entire 9,500+ TCGA-Path-Reports dataset and relying on automated (and potentially flawed) metadata extraction, the experiment now utilizes a targeted subset of 200 reports. This subset is undergoing manual annotation to establish a true “Gold Standard” ground truth.

Updated Research Questions:

1. Can open-source, 4-bit quantized models accurately deduce cancer upstaging from unstructured pathology reports?
2. Does an “Evidence-Grounded” prompt (forcing the model to extract historical/current TNM stages and quote textual evidence) improve classification accuracy compared to a baseline zero-shot prompt?
3. How does a dedicated reasoning model (Phi-4) compare to a generalist model (Qwen-2.5) in mitigating clinical hallucinations, and how frequently do standard semantic metrics (BERTScore) fail to catch these errors compared to strict text-grounding?

Part 1b: Progress in Implementation and Early Data Collection

Data Collection Progress:

The data pipeline has been successfully established. The `rosenthal/tcga-path-notes` dataset has been integrated into the Colab environment via the Hugging Face API. A representative subset of 200 reports has been extracted. Currently, the manual annotation phase is actively underway. By reading the clinical history and final diagnosis of these specific files, a manual “Upstaged: Yes/No” ground truth label is being applied to each record to guarantee the scientific validity of the final classification evaluation.

Implementation & Early Experiments:

The foundational Python infrastructure is complete and successfully tested. Because the project utilizes 7B+ parameter models on a single 16GB T4 GPU, the environment relies heavily on active memory management.

- **Quantization & VRAM Management:** A custom inference class was built using `bitsandbytes` to load models in 4-bit precision. Strict garbage collection and GPU cache clearing (`torch.cuda.empty_cache()`) were implemented. This successfully allows the sequential hot-swapping of the Qwen, Phi, and RoBERTa (BERTScore) models without triggering Out-Of-Memory CUDA crashes.
- **Pipeline Validation:** To ensure the pipeline functions prior to the completion of the manual data annotation, the codebase was tested end-to-end using a synthetically generated mock dataset. The evaluation script successfully calculates the required Macro-Averaged F1-score, runs the BERTScore comparison, and actively verifies the “Evidence Quotes.” Batch inference on the real data will commence immediately following the completion of the ground-truth labeling.